

Kartikey Kumar Singh

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LeetCode: [LeadingTheAbyss](#) (Rating: 1916)

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Codeforces: [Masochistic](#) (Rating: 1603)

EDUCATION

Walchand College of Engineering

B.Tech in Computer Science & Engineering

Sangli, Maharashtra

2024 – 2028

PD Model Sr. Sec. School

Intermediate (Class XII)

India

2022 – 2024

TECHNICAL SKILLS

- **Languages:** C, C++, Python, JavaScript
- **Frontend:** React.js, Tailwind CSS, HTML5, TypeScript
- **Backend:** Django, DRF, Django Channels, PostgreSQL, JWT Authentication
- **AI / ML:** PyTorch, TensorFlow, Keras, NumPy, Matplotlib, CNNs, ResNet, Linear & Logistic Regression
- **Core CS:** Data Structures & Algorithms, OOP, DBMS, Operating Systems

PROJECTS

TaskFlow: Scalable Notification & Task Processing System

April 2026

Python, FastAPI, Celery, Redis, PostgreSQL, Docker, React, Tailwind CSS

- **Distributed Event-Driven Architecture:** Engineered a **high-throughput microservices backend** decoupling synchronous **FastAPI** endpoints from resource-intensive workloads, effectively eliminating request timeouts and improving system availability during traffic spikes.
- **Task Orchestration and Reliability:** Integrated **Celery** and **Redis** for distributed task queuing, implementing **idempotent processing**, **exponential backoff** retries, and **Dead Letter Queues (DLQ)** to ensure guaranteed at-least-once task execution.
- **Observability and Containerization:** Developed a **React/Vite** telemetry dashboard utilizing **Recharts** for real-time monitoring of worker health and task throughput; containerized the entire stack using **Docker Compose** for seamless orchestration and environment parity.

Visual Authenticity Classifier System

March 2026

Python, PyTorch, Flask, CNN, ResNet, Grad-CAM

- **Model Development:** Trained and evaluated a **ResNet18-based binary image classifier** for distinguishing authentic and AI-generated images, achieving **95.8% validation accuracy** on a balanced dataset of **200K+ samples**.
- **Dataset Curation:** Constructed and cleaned a multi-source dataset by removing **noisy**, **low-quality**, **grayscale**, and **ambiguous** samples, improving data consistency and reducing sources of spurious learning.
- **Interpretability and Deployment:** Integrated **Grad-CAM** visualizations to inspect prediction-driving regions and deployed the model through a **Flask**-based inference interface for image upload, prediction, and qualitative analysis of model behavior.

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- **LeetCode Knight:** Reached a peak rating of **1916** (Top **4%** globally) through consistent problem-solving; completed **1000+** problems spanning DP, Graphs, and Greedy paradigms.
- **Codeforces Expert:** Attained Expert rank with a peak rating of **1603**, including a **Global Rank 501** finish in Codeforces Round 1034.
- **Global Contest Performance:** Regularly performed in the top tier of competitive programming contests, securing Global Rank under **1000** in multiple LeetCode Weekly Contests (428, 437, 439, 145).
- **IICPC 2025 & 2026:** Qualified and secured an **All India Rank under 1700** and **1800** for 2025 and 2026 respectively.
- **WCPC 2025:** Member of the winning team at the Walchand Collegiate Programming Contest, competing against multiple college teams.
- **Community Contribution:** Contributed detailed solution explanations and discussions on LeetCode, accumulating **29,000+** reputation and **1.8M+** views.

LEADERSHIP & EXTRACURRICULARS

- **Technical Clubs:** Active Member of **WCE ACM Student Chapter** and **Google Developers Student Club (GDSC)** (2024–Present).
- **CodeChef Chapter:** Member of WCE CodeChef Chapter; organized peer-learning sessions on C++ and DSA.